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Metallurgical microstructure classification using CNN: A comprehensive study on heat treatment analysis for steel

N.P. Wankhade [a], V.P. Sale [a], R.S. Yadav [a], P.C. Jikar [a,*], S.R. Gadgekar [a], N.B. Dhokey [b]

[a] School of Mechanical Engineering, MIT Academy of Engineering, Alandi, Pune 412105, India
[b] Department of Metallurgy and Material Science, COEP Technological University, Wellesley Rd, Shivajinagar, Pune, Maharashtra 411005, India

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ABSTRACT

Predicting metallurgical microstructures is challenging due to the intricate features that emerge during manufacturing. This research utilizes Convolutional Neural Networks (CNNs) to recognize microstructure images of mild steel alloys, representing diverse microstructures. The study focuses on forecasting steel microstructure properties, specifically identifying and predicting Pearlite (P) and Bainite (B) microstructures after steel heat treatment, using a CNN approach. This comprehensive workflow leads to robust microstructure classification, achieving a model training accuracy rate of 98.95 %. While its final validation accuracy of 98.8 %, highlighting its capability to generate precise predictions across both classes seamlessly. Valuable insights from confusion matrices highlight the model's performance in distinguishing between "Bainite" and "Pearlite" microstructures. Additionally, the study underscores the significance of CNNs in metallurgical microstructure analysis, with insightful confusion matrices revealing the model's capability to differentiate between microstructures. This is useful for innovative microstructure analysis in materials science, emphasizing the transformative potential of AI in enhancing our understanding of processing-microstructure relationships in industrial applications.

1. Introduction

Predicting metallurgical microstructures is inherently challenging due to the complex structures and diverse features that develop during the metallurgical processes. Study incorporates the use of Convolutional Neural Networks (CNNs) to recognize microstructure images of alloys of Aluminum (Al) and Iron (Fe), based on their various morphological structures, including spheres, petals, needles, and dendrites. The CNN achieved an impressive accuracy rate of 94.6 % [1]. Recently, several metal manufacturing industries have expressed interest in use of computer vision and image processing into their operations. This technology enables the cost-effective design of materials to achieve desired qualities. Microstructure modeling can be particularly valuable in implementing various processes, including heat treatment. Study focuses on predicting the microstructure properties of phases during the heat treatment of steels [2]. Moreover an automated microstructure algorithm utilized to assess the quality of materials. In some cases, computational image processing techniques have proven to be more effective than traditional methods for microstructures [3]. The study predicts that microstructure depends on mechanical properties in additively manufactured SS316L using a CNN algorithm. Among all the data-driven models tested, the CNN provides the best prediction with minimal preprocessing time. The study also provides a summary of current research on various aspects of phase transformation in steel and its improved properties. To explore diffusion-controlled phase transformation in steel and the kinetics of austenitization in pearlite and ferrite transformation in low-carbon steel. Identifying such complex phase transformations is a challenging task that requires expertise to understanding microstructures accurately [4,5]. Additionally an adaptive characterization of a titanium alloy (Ti-6Al-4 V) microstructure dataset was conducted using a two-stage machine learning approach. The study employed quantitative data from micrographs for microstructure feature recognition using a CNN algorithm. It utilized a dataset comprising 1225 images of the alloy and reported an accuracy of approximately 93 % with a margin of ± 1.17 % [6]. Furthermore pattern recognition of optical microscopic images of steel was performed using the random forest statistical algorithm technique. The author compared the accuracy of microstructure percentage with linear analysis [7]. Additionally, the carbon content present in iron (Fe) influences the categorization of steel into three significant types: low carbon, medium carbon, and high carbon steel. These distinctions are determined by various heat treatments such as annealing, normalizing, tempering, and hardening, which induce

* Corresponding author.
E-mail address: prashant.jikar@yahoo.com (P.C. Jikar).

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different microstructures at diverse temperatures and cooling rates. These microstructures, including Martensite (M), Pearlite (P), Bainite (B), and Spheroidite (S), are observable in the Time-Temperature-Transformation (TTT) diagram [8]. Depending on the cooling rates, steel can exhibit various microstructures, including Ferrite (F), Pearlite (P), Bainite (B), and Martensite (M). These transformations stem from both displacive and reconstructive processes within the Austenite crystal structure (A). Moreover, cementite precipitation occurs at varying diffusion rates, except in the case of Martensite (M). These transformations and precipitation patterns are depicted on the Continuous Cooling Transformation (CCT) diagram. The study employed a dataset comprising 1225 images, from which multiple morphological features were extracted using algorithmic operations. A Convolutional Neural Network (CNN) was trained using the Keras API, consisting of three layers to classify microstructures into three categories: laminar, duplex, and acicular. Data analysis focused on critical aspects such as phase identification, grain characterization, and defect detection, leveraging advanced image processing techniques including phase segmentation, Sobel, watershed, and region-based segmentation. Integration of CNN into the algorithmic approach enhanced its capabilities and improved the accuracy of microstructural analysis. In a Machine Learning (ML) approach, a dataset of 730 images was utilized for training purposes [9]. The study offers insights into computer vision and ML techniques for microstructural characterization and analysis, highlighting their importance in microstructural science. Traditionally, human judgment guided measurement and method selection, but recent technological advancements offer new avenues for extracting information from microstructures. CNNs, particularly in the context of metallurgical microstructures, are suggested for high-dimensional image representation, aiding in understanding the relationship between processing and microstructure properties [10]. Focused on GCr15 bearing steel, a high-carbon alloy enriched with chromium and manganese, the research analyzed 58 scanning electron microscope (SEM) images. The Mask R-CNN algorithm was employed to extract and analyze microstructural features, utilizing data from the COCO dataset, a recognized resource in computer vision and object detection tasks. This approach facilitates precise characterization and comprehension of the microstructure in high-quality GCr15 bearing steel alloy [11]. The study emphasizes the need for standardization of methods, enhancement of data quality and quantity, generalization of models, to explain of AI decisions, real-time analysis, multimodal data integration, and practical implementation of AI in industrial settings. To use CNN, a deep learning model, the research accurately categorizes various phases of microstructure, claiming high accuracy in microstructure classification [12]. In AI and ML the CNN stands for Convolutional Neural Network, a specialized type of deep learning algorithm designed for tasks such as image processing, computer vision, and pattern recognition, with particular proficiency in image classification and object detection [13]. The utilization of the TensorFlow library involves constructing and training CNN models to classify images into predefined classes. TensorFlow, in tandem with Keras, is a widely used open-source machine learning library developed by Google for building and training deep learning models [14,15]. Researchers widely acknowledge the advantages of image processing in tasks such as identifying grain boundaries, grain sizes, and grain segmentation [16]. From the literature review, it was evident that many studies have concentrated on microstructure prediction for alloys like steel (SS316L), Ti, Fe-Ga, and high-ultra carbon, whereas very few have addressed microstructure prediction for low carbon plain steel. The literature also highlights the widespread recognition and acceptance of the CNN algorithm within the research community for microstructure prediction. This study introduces a novel approach to phase identification and prediction for low carbon steel microstructures using the CNN algorithm, bridging an existing research gap and presenting an opportunity for further advancement in the field of metallurgical microstructure analysis. The study proposes a phase identification technique to predict bainite and pearlite microstructure images, utilizing a total of 2508 images of mild steel.

2. Methodology

The workflow of the proposed methodology involves various stages, including sample preparation, heat treatment, microstructure extraction, dataset formation, CNN architecture building, and finally, model verification and testing.

2.1. Sample preparation and heat treatment

In this research, low carbon steel, specifically mild steel, was chosen as the material of interest. To ensure high accuracy, a dataset comprising microstructure images of both good quantity and quality was assembled. Initially, mild steel rod samples were obtained and cut into small pieces of uniform size. The selected dimensions for the samples were a length & diameter of 30 mm and 20 mm respectively. These samples were then cleaned thoroughly to remove any dirt or contaminants. Finally, the samples were subjected to a heat treatment process. The heat treatment involved three main phase: annealing, normalizing and quenching. For clarity in subsequent discussions, a nomenclature system was adopted for the treated samples. The nomenclature includes, Sample MS_A refers to the sample subjected to slow cooling in the furnace for 24 h. Sample MS_B1 and MS_B2 denote the two samples rapidly cooled by immersion in quenching media water and oil, respectively. Sample MS_C: Represents the sample cooled by normalizing in air. In the Heat treatment process all samples were initially heated to a temperature of $925 \pm 25\ ^{\circ}C$ using a muffle furnace. This temperature falls within the Austenitic temperature zone and facilitated recrystallization and grain growth within the material. Fig. 1 illustrates the heat-treated samples along with their corresponding cooling techniques for reference.

2.2. Microstructure extraction

After completing the heat treatment and cooling process, the samples underwent a preparation procedure to facilitate microstructure analysis. During surface polishing, abrasive papers with varying grit sizes, ranging from 100 to 2000, were used to polish the samples. This process aimed to achieve a smooth and uniform surface suitable for microscopy analysis. To enhance the microstructure further after achieving a smooth surface finish, the samples were exposed to an etchant to corrode grain boundaries within the microstructure. This step was crucial for making the microstructure visible under a microscope. Subsequently, the

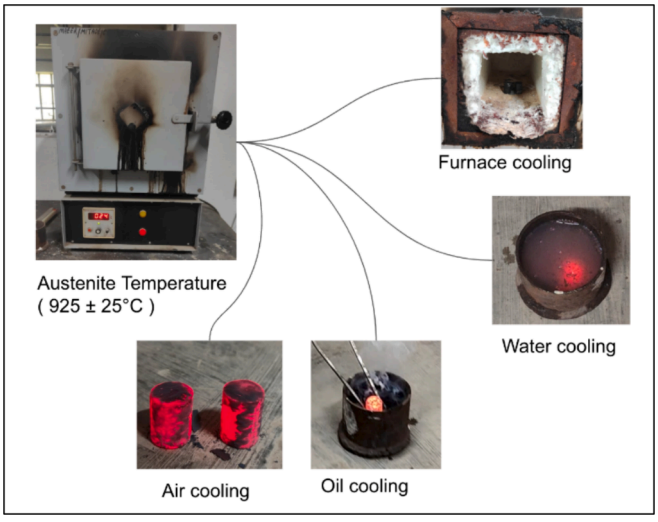


Fig. 1. Heat treated mild steel sample subjected to different cooling and quenching techniques.

samples microstructure prepared for analysis. A total of five samples were prepared to ensure comprehensive microstructural observation and analysis. The process of image extraction from the heat-treated samples was carried out using an Optical Inverted Microscopy setup, providing the necessary visual data for subsequent analysis and characterization.

2.3. Dataset preparation

In the image sectioning process, the original images, with dimensions of 1944 × 2592 pixels as shown in Fig. 2, were divided into smaller, non-overlapping sections of size 388 × 432 pixels as depicted in Fig. 3. This segmentation was carried out to ensure clear annotations and to enable the classifier to train on features specific to the annotated label. It ensured that there was no spatial overlap between any two sectioned in images.Fig. 4.

In the dataset composition the dataset employed in this study comprised a total of 2508 images, categorized into two distinct classes. To establish the training and testing data, 80 % of the images were allocated for training, while the remaining 20 % were reserved for testing. The first class consisted of images obtained from air-cooled samples (MS_C), while the second class comprised images from oil-quenched samples (MS_B2). A subset of 2508 images was selected for the training process, and within this subset, 20 % of the images were further designated as the test dataset.

In the pixel value scaling as a common practice in many image processing and machine learning tasks, the pixel values were scaled to a range between 0 and 1. This scaling ensures better numerical stability and enhanced convergence during training. It is particularly beneficial for machine learning algorithms that operate optimally with input values within a bounded range. In preprocessing stage after scaling, preprocessing tasks were applied to efficiently process and manipulate the data, especially when working with batched or iterated data for training or evaluation. This step is particularly useful when using frameworks like Tensor Flow, where datasets are a common data structure for machine learning tasks. The data splitting of the dataset was divided into three subsets: 70 % of the data was allocated for training, 20 % for validation, and 10 % for testing. An additional 1 was added to ensure that the validation set contained at least one data point, and the same was done for the testing set to a minimum size for it. This practice is common when partitioning a dataset into subsets for training, validation, and testing, ensuring a robust evaluation of the model's performance.

2.4. Noise reduction and edge detection

2.4.1. Noise reduction

Since edge detection is susceptible to noise in the image, first step is to remove the noise in the image with a 5x5 Gaussian filter. This helps to remove high-frequency noise in the image, resulting in smother gradients for edge detection

Sample of 2D Gaussion function calculation as given below:

$$G_\sigma(x,y) = \frac{1}{2\pi\sigma^2} e^{\frac{-(x^2+y^2)}{2\sigma^2}}$$

2.4.2. Gradient calculation

Gradient calculation is a fundamental operation in image processing and computer vision. Gradients represent the rate of change of pixel intensities in an image and are essential for various tasks, such as edge detection and feature extraction. The Sobel operator is used to calculate the first derivative of an image in both the x-direction (horizontal gradient) and the y-direction (vertical gradient). This allows to detect edges and changes in intensity in an image. The Sobel x and Sobel y operators are represented by 3x3 convolution kernels are shown in Fig. 8, and they provide gradient information in two perpendicular directions. In Fig. 6, x-axis denote the magnitude increasing at right direction where as y axis denotes in the down direction. The resultant gradient magnitude can be measured as follows:

$$Edge_Gradient(G) = \sqrt{G_x^2 + G_y^2}$$

In addition, gradient direction calculation as given below:

$$Angle(\theta) = tan^{-1}\left(\frac{G_y}{G_x}\right)$$

After applying Sobel edge detection algorithm in our input dataset [from Fig. 5(b)] we get G_x as shown in Fig. 8(a) and G_y as shown in Fig. 8 (c) after using Sobel edge detection. Compare the Air and Oil input dataset with help of Sobel Edge Detection.Fig. 7

Other edge detection techniques such as canny and Laplacian are also tested on our data set as shown in Fig. 9. It is observed that Sobel edge detection takes very less computational time over our data set, and one more important feature in Sobel operator is that it is simple as it uses the first derivative filter. Hence, the chances of feature loss during edge detection are few.

Fig. 9 illustrates the testing of other edge detection methods on our data set, including canny and laplacian. In our data set, it is seen that Sobel edge detection requires relatively little computational time. Another crucial aspect of the Sobel operator is that it is straightforward because it makes use of the first derivative filter. Therefore, there is a

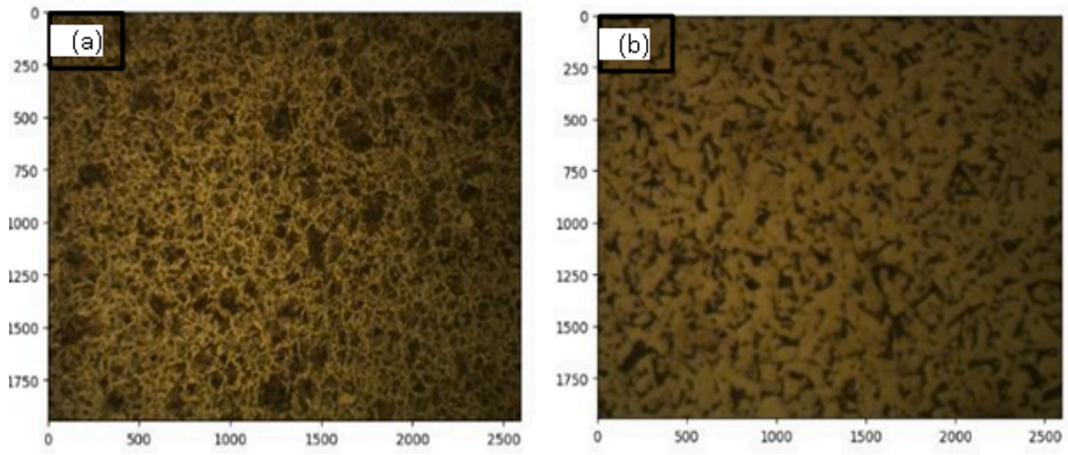


Fig. 2. Image of (a) oil quenched and (b) air cooled (Normalizing) samples obtained before a sectioning.

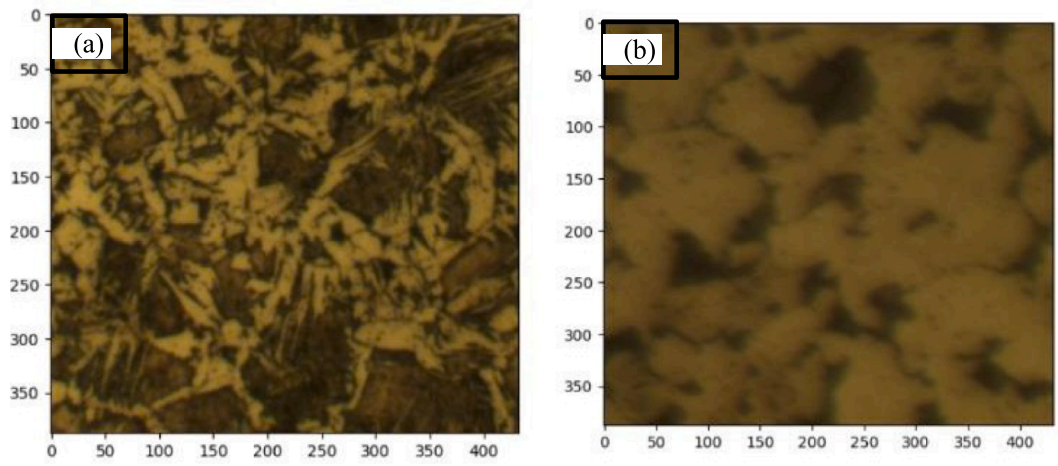


Fig. 3. Image obtained after sectioning the original images (1944 × 2592 pixels) into smaller sections of size (388 × 432 pixels) (a) Oil patched (b) Air patched.

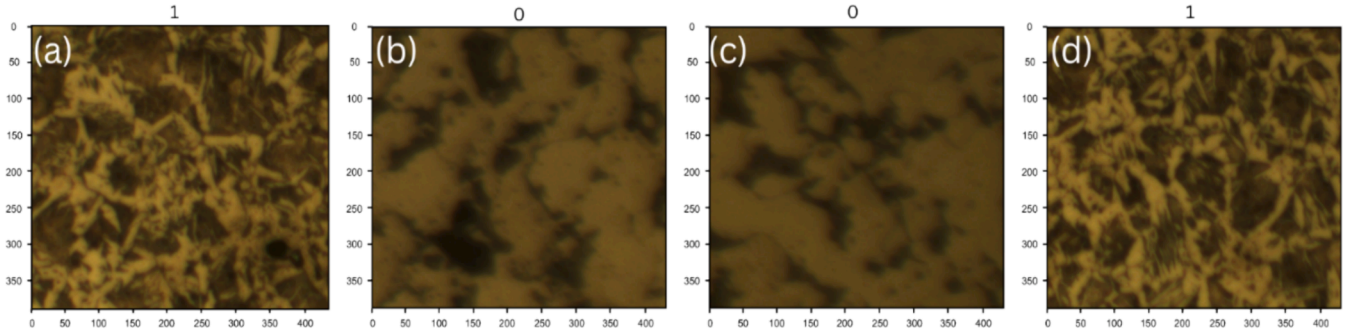


Fig. 4. Image of pixel values are converted to floating-point numbers (a) 1 patched air; (b) 0 patched oil it is represented in arrays; (c) 0 patched oil and (d) 1 patched air it is represented in arrays.

low likelihood of feature loss during edge identification. The output of the laplacian edge detection technique, which highlights edges but may also contain some noise, is shown in Fig. 9(a). After using Canny edge detection, the output in Fig. 9(b) is a clear and accurate edge map, which typically contains the strongest and most pertinent edges.

2.4.3. *Image binarization*

In this study, Otsu's algorithm is used to image binarization. Otsu's algorithm is a valuable tool in image processing and computer vision, as it simplifies the task of determining an appropriate threshold and is well-suited to a wide range of applications. In our dataset for Canny edge detection we apply 'Otsu's algorithm'.

Otsu's thresholding, where the threshold value is determined automatically based on the image's histogram. These techniques are essential in various image processing applications, such as object detection, image segmentation, and feature extraction.

2.5. *Software detail*

In CNN techniques, various software tools and libraries are commonly used to implement, train, and evaluate neural networks. Some of the key software components frequently used in CNN techniques include Python, Deep Learning Frameworks: Tensor Flow, Keras, PyTorch, Open CV, Numpy, Pandas and Jupyter Notebook along with various activation function. Algorithm used in our study is Gaussian blur, sobel edge detection, laplacian edge detection, canny edge detection, otsu's thresholding.

2.6. *CNN architecture building microstructure image classification*

Subsequently the image preparation for the CNN processing employed to trained and predict microstructure images between two classes. After preprocessing and denoising microstructure images CNN model is prepare to predict between two classes of microstructure Pearlite (P) and Bainite (B) images (Fig. 10).

In the CNN architecture network employs essential components like Conv2D for convolution, MaxPooling 2D for pooling, flatten for data reshaping, and dense for neural network connections. The hidden layer has 128 nodes and uses the "ReLU" activation function, introducing non-linearity to the network. The primary goal is binary classification of images into two classes based on phases. An image with intensity variance is classified as "1," while others are labeled "0." Since it's a binary classification task, the final layer employs the sigmoid activation function to produce the classification output. In the layers overview total three convolution layer of with 16 and 32 filters with kernel size: 3x3 employed by using ReLU activation function. The initial convolutional layer serves as the feature extractor. Following each convolutional layer, a max-pooling layer is applied for 2D microstructure image. This layer down samples the spatial dimensions of the feature maps while retaining crucial features and reduces spatial dimensions. In layer flatten the feature maps into a 1D vector, preparing them for input into the fully connected layers. A fully connected (dense) layer with 256 units and a ReLU activation function. This layer learns higher-level representations from the flattened feature vectors. Another dense layer comprises a single unit and uses a sigmoid activation function. This configuration is suitable for binary classification tasks, such as distinguishing between Pearlite (P) and Bainite (B) microstructures.

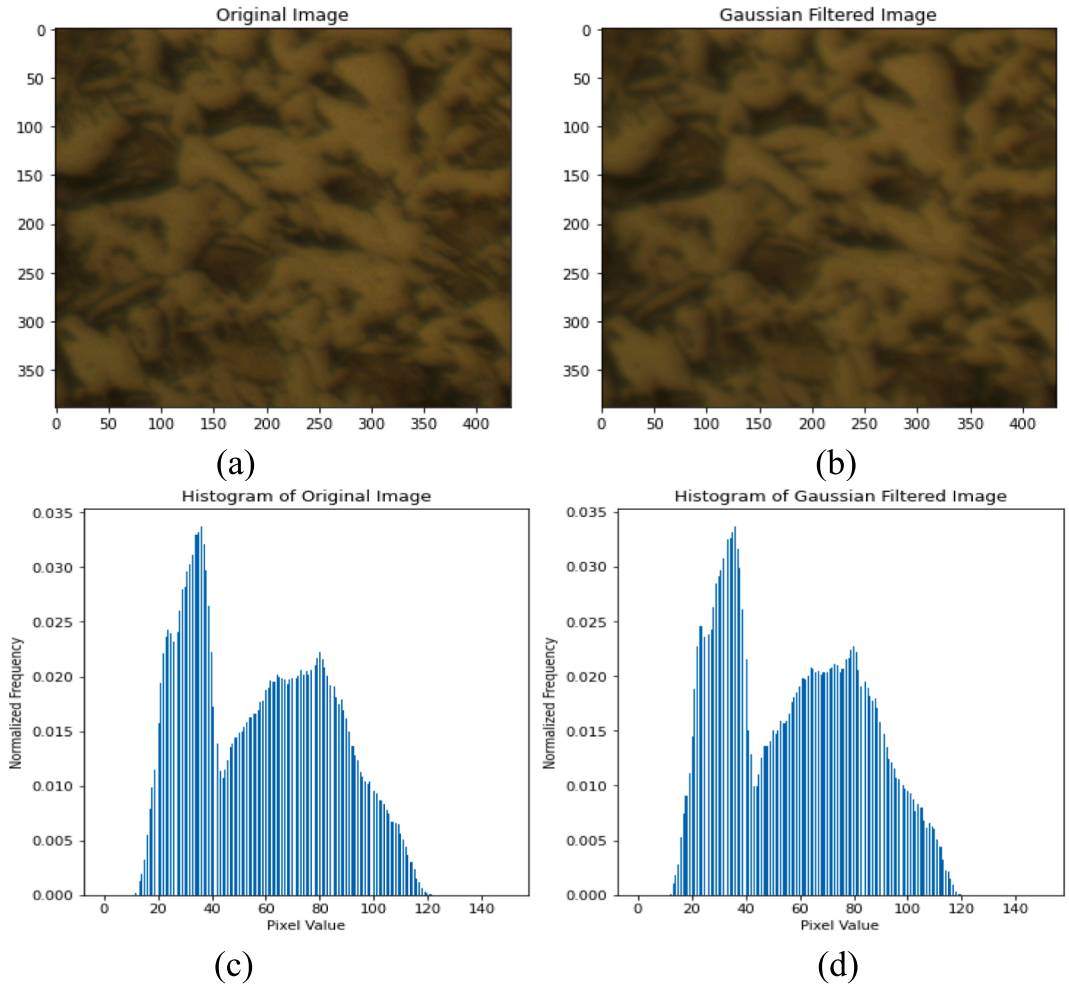


Fig. 5. Oil Image (a) input image (before denoising) (c) corresponding histogram of (a); (b) Input Image [after Gaussian Filtered from input image (a)] and (d) Corresponding Histogram of (b);

$$\begin{matrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{matrix} \qquad \begin{matrix} +1 & +2 & +1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{matrix}$$

Gx **Gy**

Fig. 6. Kernal Matrix.

The provided CNN architecture is designed to effectively process microstructure images, extract relevant features, and make binary classification predictions. It incorporates convolutional layers for feature extraction and max-pooling layers for spatial down sampling. The fully connected layers further learn abstract representations, ultimately leading to accurate microstructure classification.

The model consists of a sequence of convolutional and pooling layers, followed by fully connected layers.

3. Result and discussion

In this study, we embarked on the task of binary classification of microstructure images, specifically aiming to distinguish between two fundamental phases: bainite and pearlite. Through meticulous experimentation and analysis, we uncovered a series of insightful outcomes that shed light on the efficacy and performance of our classification model.

3.1. Confusion matrix

Our first step in evaluating the model's performance was through the construction and analysis of the confusion matrix. This matrix provided a comprehensive breakdown of the model's predictions against the ground truth labels. Each quadrant of the matrix—True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN)—offered critical insights into the model's ability to accurately discriminate between bainite and pearlite phases (Fig. 11) (Table 1).

3.2. Training performance

The training accuracy, standing at an impressive 98.95 %, and the corresponding training loss of 34.1, offer a glimpse into the model's performance during the training phase. Training accuracy elucidates the proportion of correctly classified instances during training iterations, offering insights into the model's learning dynamics and convergence. Conversely, training loss signifies the model's optimization process, reflecting the degree of discrepancy between predicted and actual labels during training. Analyzing these metrics allows for a deeper understanding of the model's learning trajectory and optimization behavior (Fig. 12).

3.3. Validation performance

Validation accuracy, quantified at an impressive 98.8 %, serves as a crucial metric in assessing the model's overall performance on unseen

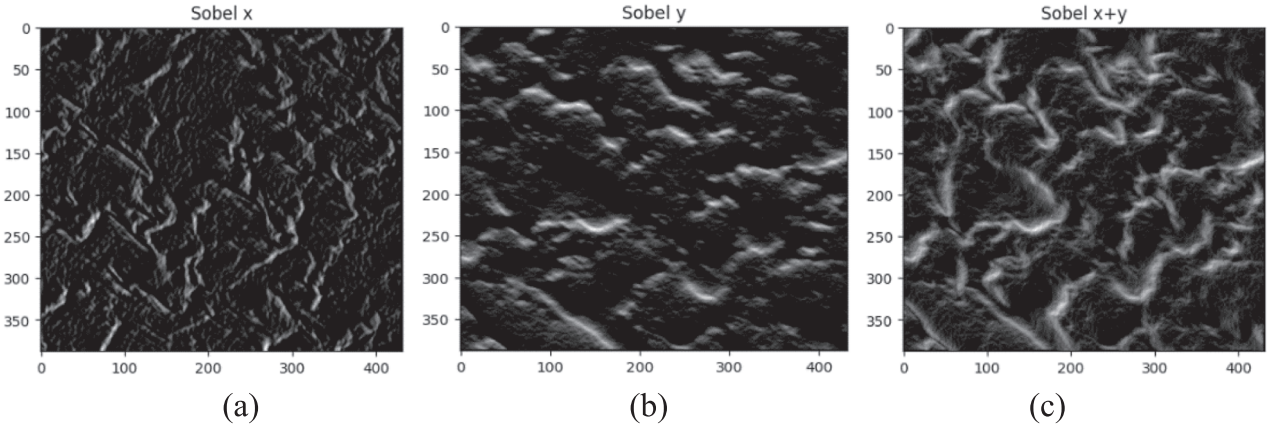


Fig. 7. Air Image (a) Sobel x (b) Sobel y (c) After Sobel x + y Edge detection resulting gradient magnitude G.

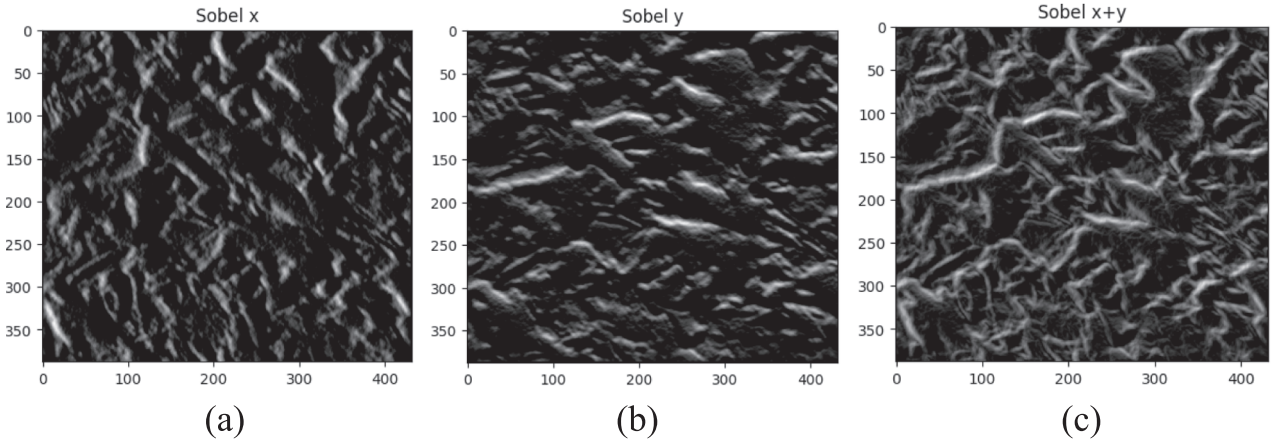


Fig. 8. Oil Image (a) Sobel x (b) Sobel y (c) After Sobel x + y Edge detection resulting gradient magnitude G.

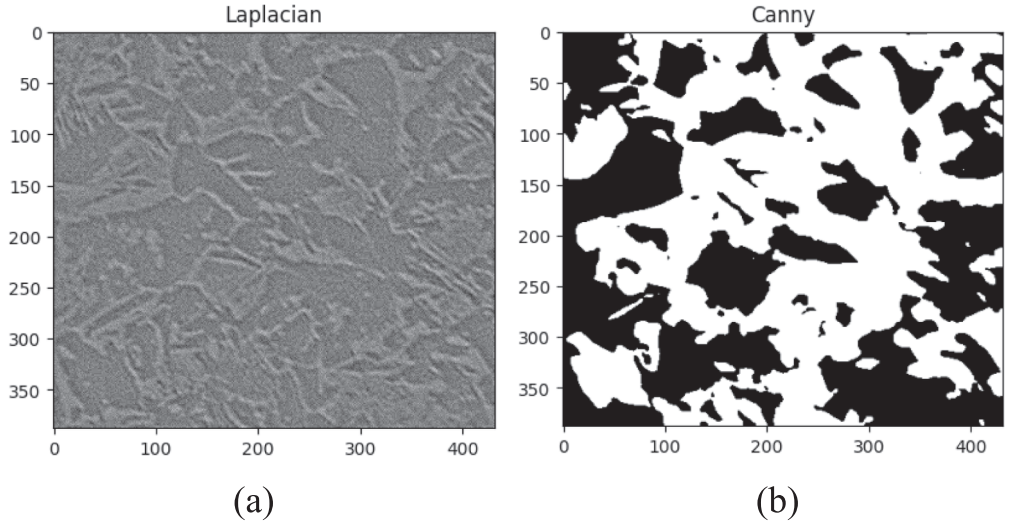


Fig. 9. Oil Images (a) After Laplacian edge detection and (b) After Canny edge detection.

data. It represents the proportion of correctly classified instances out of the total validation dataset. Concurrently, validation loss, recorded at 59.5, indicates the model's degree of error during the validation phase. These metrics collectively offer a snapshot of the model's generalization ability and effectiveness in handling unseen data, providing vital information on its robustness and reliability (Fig. 13).

4. Conclusion

In the present research work Convolutional Neural Networks (CNNs)

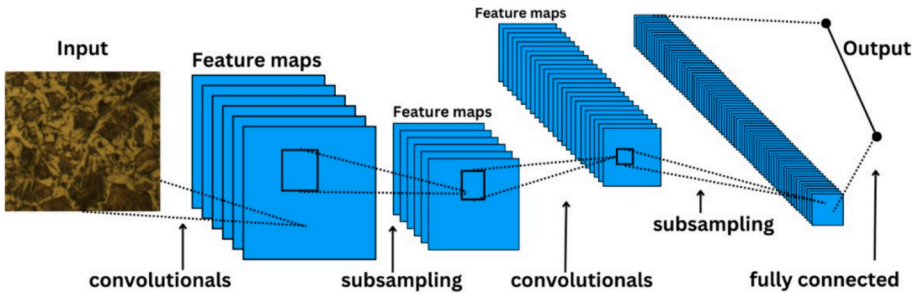


Fig. 10. Convolutional Neutral Network (CNN) architecture for microstructure prediction.

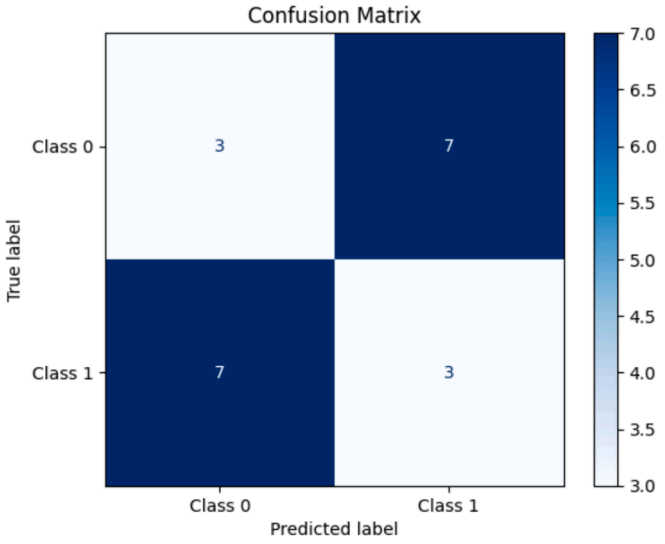


Fig. 11. Confusion Matrix.

Table 1
Training and Validation Results.

Epoch	Training Loss	Training Accuracy	Validation Loss	Validation Accuracy
1	32.08	84.49 %	26.01	95 %
2	24.82	89.23 %	143.42	50 %
3	12.95	95.02 %	28.40	80 %
4	11.38	94.98 %	22.79	90 %
5	6.86	97.75 %	3.14	100 %
6	28.41	89.19 %	34.20	70 %
7	9.18	96.78 %	53.77	70 %
8	5.82	98.23 %	69.92	70 %
9	7.48	97.07 %	4.67	100 %
10	3.41	98.95 %	59.5	98.98 %

employed for the prediction and classification of metallurgical microstructures, particularly for low carbon steel.

(a) Study observe the training accuracy rate of 98.95 % the model effectively captured the intricate characteristics and patterns inherent in the training data. To generalize evidenced by its final validation accuracy of 98.8 %, highlighting its capability to generate precise predictions across both classes seamlessly.
(b) The model's adeptness at adaptively tuning its parameters during training led to significant improvements in accuracy for both the training and validation sets. Variations in loss and accuracy metrics throughout training are expected, reflecting the model's ongoing learning process and its continual adjustments to the dataset.
(c) Furthermore, the significance of microstructure modeling, especially in processes like heat treatment. Study focus on predicting microstructure properties during heat treatment of steels shed light on the critical role of computational image processing techniques in complementing traditional methods for comprehending microstructures.
(d) The integration of CNNs into methodology showcased its prowess in microstructure image classification. The CNN architecture, with its convolutional layers, max-pooling layers, and fully connected layers, effectively processed microstructure images, enabling accurate binary classification of phases such as Pearlite (P), and Bainite (B).
(e) The valuable insights provided by confusion matrices, highlighting the model's performance in distinguishing between "Bainite" and "Pearlite" microstructures.

In conclusion, present study bridged the gap in microstructure analysis, particularly for low carbon steel, and introduced an innovative approach to phase identification and prediction using CNN algorithms. Study emphasized the importance of CNNs in materials science explored their potential in enhancing understanding of the intricate relationship between processing and microstructure properties. As we look ahead, standardization of methods, data quality improvement, and the practical implementation of AI in industrial settings will continue to shape the future of metallurgical microstructure analysis.

5. Future scope

The future research work involve improving dataset quality, quantity and diversity through collaborations with industry partners or utilizing advanced data augmentation techniques to enhance model robustness and generalization. Additionally, exploring techniques for incorporating domain knowledge or expert guidance into the CNN architecture to improve interpretability and help identify relevant microstructural features. Furthermore, investigating more efficient model architectures or optimization strategies to reduce computational requirements without sacrificing performance to enhance scalability and real world applicability. Finally, exploring interdisciplinary collaborations with experts in materials science, computer vision, and machine learning to foster innovation and drive advancements in metallurgical microstructure analysis using CNNs.

CRediT authorship contribution statement

N.P. Wankhade: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **V.P. Sale:** Writing – original draft, Visualization, Software, Data curation. **R.S. Yadav:** Software, Methodology, Data curation. **P.C. Jikar:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **S.R. Gadgekar:** Visualization. **N.B. Dhokey:** Writing – review & editing.

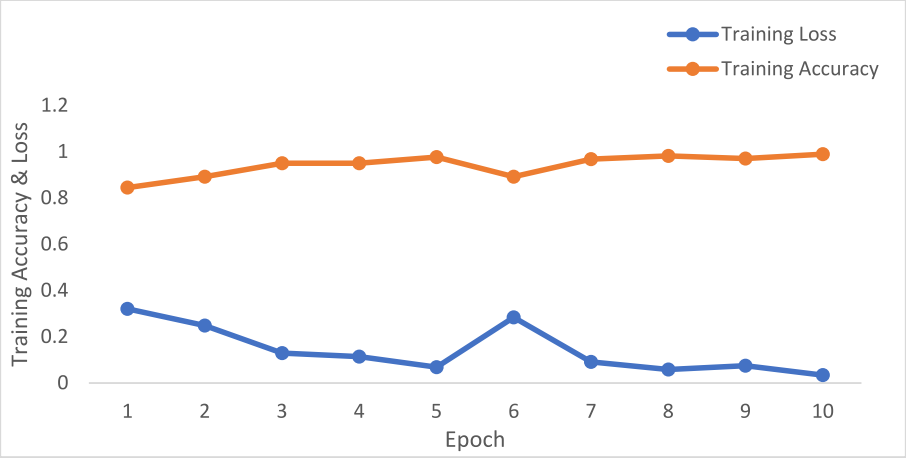


Fig. 12. Training Accuracy & Loss Vs Epoch.

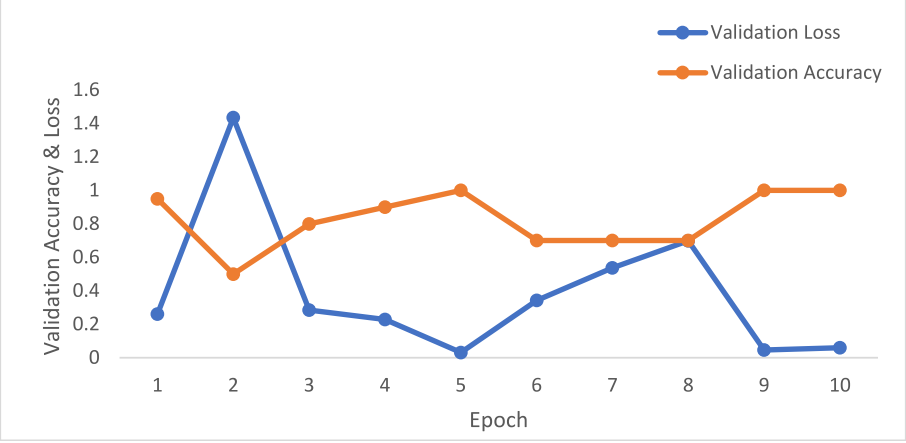


Fig. 13. Validation Accuracy & Loss Vs Epoch.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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